1 $$\frac{\partial f}{\partial x} = \frac{\ln(x^2-y)}{2\sqrt{x}} + \frac{2x^{3/2}}{x^2-y} \Rightarrow \frac{\partial f}{\partial x}(1,0) = 2$$

$$\frac{\partial f}{\partial y} = -\frac{\sqrt{x}}{x^2-y} \Rightarrow \frac{\partial f}{\partial y}(1,0) = -1$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\frac{4x^{3/2}}{x^2-y} - \frac{1}{\sqrt{x}}\ln(x^2-y)}{4x} + \frac{3\sqrt{x}(x^2-y) - 4x^{5/2}}{(x^2-y)^2}$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2}(1,0) = \frac{4}{4} - 1 = 0$$

$$\frac{\partial^2 f}{\partial y \partial x} = -\frac{1}{2\sqrt{x}(x^2-y)} + \frac{2x^{3/2}}{(x^2-y)^2} \Rightarrow \frac{\partial^2 f}{\partial y \partial x}(1,0) = \frac{3}{2}$$

$$\frac{\partial^2 f}{\partial y^2} = -\frac{\sqrt{x}}{(x^2-y)^2} \Rightarrow \frac{\partial^2 f}{\partial y^2}(1,0) = -1$$

$$\Rightarrow T_2(x,y) = 2(x-1) - y + \frac{1}{2}\left[3(x-1)y - y^2\right]$$

$$f\left(\frac{11}{10}, -\frac{1}{10}\right) \approx T_2\left(\frac{11}{10}, -\frac{1}{10}\right) = \frac{2}{10} + \frac{1}{10} + \frac{1}{2}\left[-\frac{3}{100} - \frac{1}{100}\right]$$

$$= \frac{30}{100} - \frac{2}{100} = \frac{7}{25}$$

② (i) $(x,y) \in \operatorname{int} M$ ... $\begin{aligned} 2x+1&=0 \\ 2y&=0 \end{aligned} \iff x=-\frac{1}{2},\ y=0$

Bod $(-\frac{1}{2},0) \in \operatorname{int} M$ je podezřelý z extrému.

(ii) $(x,y) \in \partial M$ ...

$$2x+1+\lambda\frac{x}{2}=0$$
$$2y+\lambda y=0 \quad \ldots \quad y(2+\lambda)=0$$
$$\frac{x^2}{4}+\frac{y^2}{2}=1$$

- $y=0 \Rightarrow x=\pm 2$
- $y\neq 0$. Pak $\lambda=-2$. Z 1. rovnice plyne, že $x+1=0$, tj. $x=-1$. Tedy $y=\pm\sqrt{\frac{3}{2}}$.

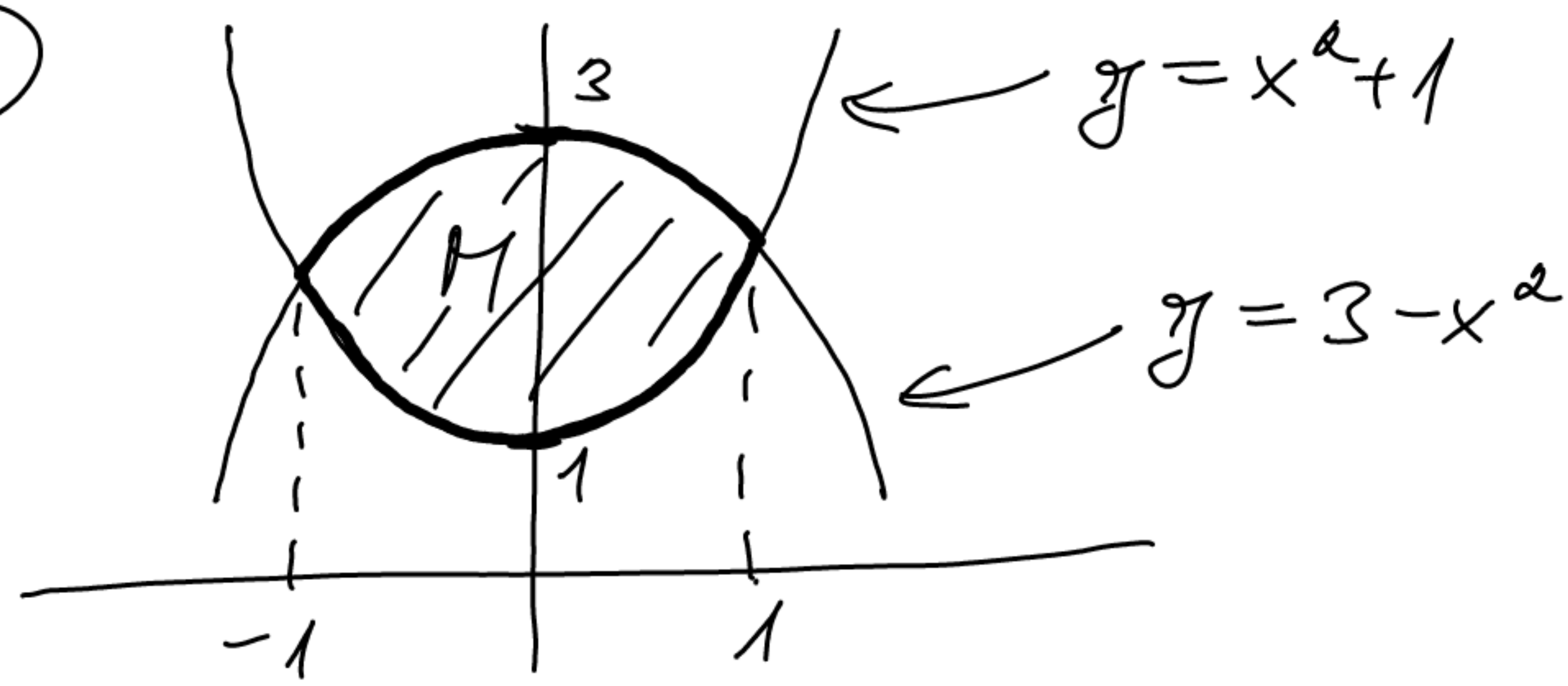
Podezřelé body z extrému ležící na ∂M jsou $(2,0)$, $(-2,0)$, $(-1,\sqrt{\frac{3}{2}})$, $(-1,-\sqrt{\frac{3}{2}})$.

Protože

$f(2,0)=6$, $f(-2,0)=2$, $f(-1,\pm\sqrt{\frac{3}{2}})=\frac{3}{2}$,

$f(-\frac{1}{2},0)=-\frac{1}{4}$, je $(2,0)$ bod maxima f na M a $(-\frac{1}{2},0)$ bod minima f na M.

③



$$\int_M y e^{x^3-3x}\,d\lambda_2 = \int_{-1}^{1}\int_{x^2+1}^{3-x^2} y e^{x^3-3x}\,dy\,dx$$

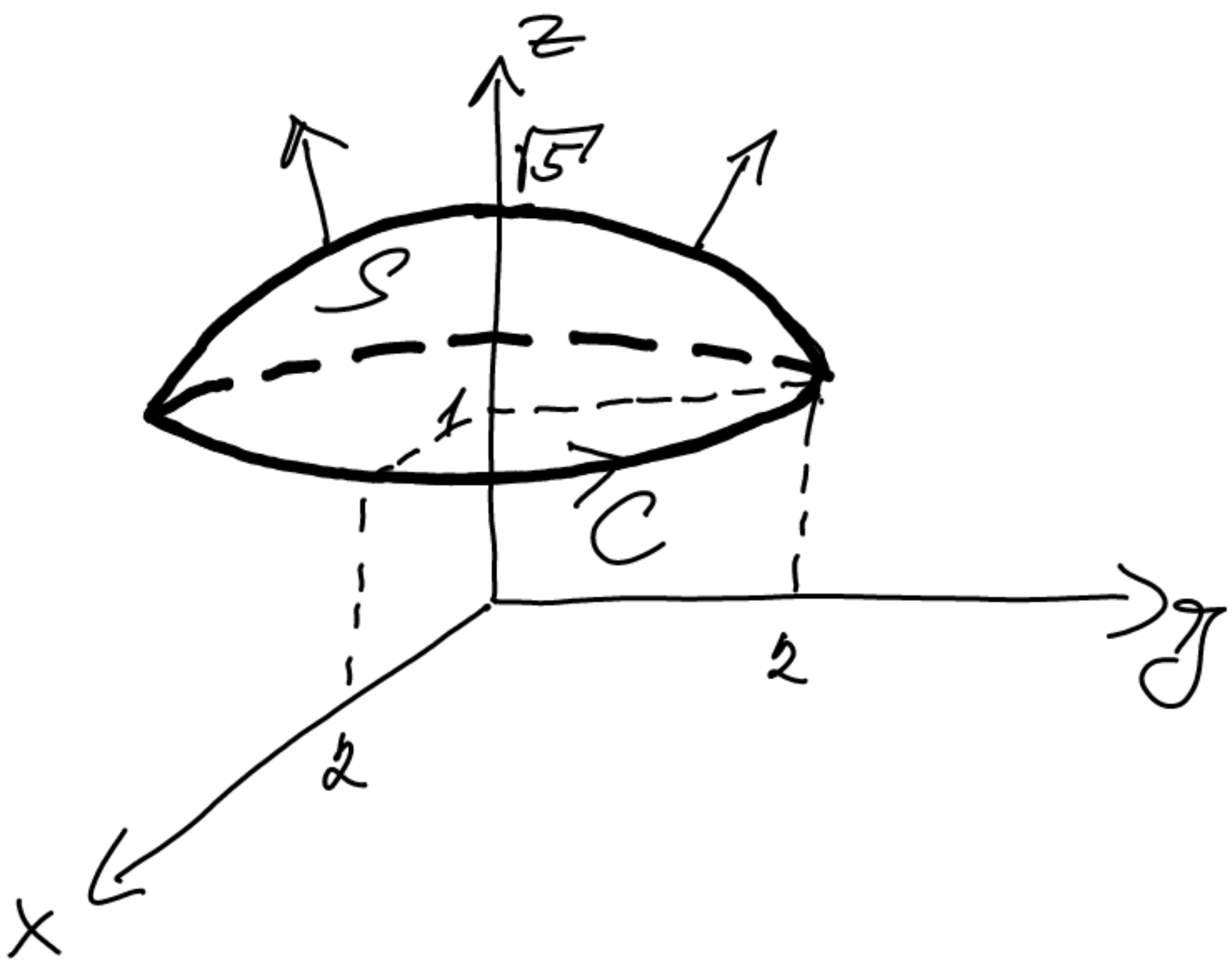
$$= \int_{-1}^{1} \frac{1}{2}\left[(3-x^2)^2-(x^2+1)^2\right] e^{x^3-3x}\,dx$$

$$= \frac{1}{2}\int_{-1}^{1}\left(9-6x^2+x^4-x^4-2x^2-1\right) e^{x^3-3x}\,dx$$

$$= \frac{1}{2}\int_{-1}^{1}\left(8-8x^2\right) e^{x^3-3x}\,dx = \left|\begin{array}{l} u=x^3-3x \\ du=(3x^2-3x)dx \\ 1 \rightsquigarrow -2 \\ -1 \rightsquigarrow 2 \end{array}\right|$$

$$= 4\int_{2}^{-2} -\frac{1}{3} e^{u}\,du = \frac{4}{3}\int_{-2}^{2} e^{u}\,du = \frac{4}{3}\left(e^{2}-e^{-2}\right)$$

(4)



$$\varphi(t) = (2\cos t, 2\sin t, 1) \quad , \; t\in[0, 2\pi]$$

$$\varphi'(t) = (-2\sin t, 2\cos t, 0)$$

$$\int_S \nabla\times F \bullet d\sigma = \int_C F \bullet ds$$

$$= \int_0^{2\pi} (2\sin t, \cos t, e^{-2}) \bullet (-2\sin t, 2\cos t, 0)\, dt$$

$$= \int_0^{2\pi} -4\sin^2 t + 2\cos^2 t \, dt = \int_0^{2\pi} -4 + 6\cos^2 t \, dt$$

$$= -8\pi + 6\int_0^{2\pi} \frac{1+\cos(2t)}{2}\, dt = -8\pi + 6\pi$$

$$= -2\pi$$

(5) $f(x)=\frac{x+5}{(x-3)(x+1)}=\frac{2}{x-3}+\frac{-1}{x+1}$

$$\frac{1}{x-3}=\frac{1}{(x-1)-2}=-\frac{1}{2}\frac{1}{1-\frac{x-1}{2}}=-\frac{1}{2}\sum_{n=0}^{\infty}\frac{(x-1)^n}{2^n}$$

pro $|x-1|<2$

$$\frac{1}{x+1}=\frac{1}{(x-1)+2}=\frac{1}{2}\frac{1}{1-\left[-\frac{x-1}{2}\right]}=\frac{1}{2}\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n}(x-1)^n$$

pro $|x-1|<2$.

Tedy

$$f(x)=\sum_{n=0}^{\infty}-\frac{(x-1)^n}{2^n}-\sum_{n=0}^{\infty}\frac{(-1)^n}{2^{n+1}}(x-1)^n$$

$$=\sum_{n=0}^{\infty}\frac{-2-(-1)^n}{2^{n+1}}(x-1)^n \quad \text{pro } |x-1|<2.$$

(Polom. konvergence je tedy $R=2$.)